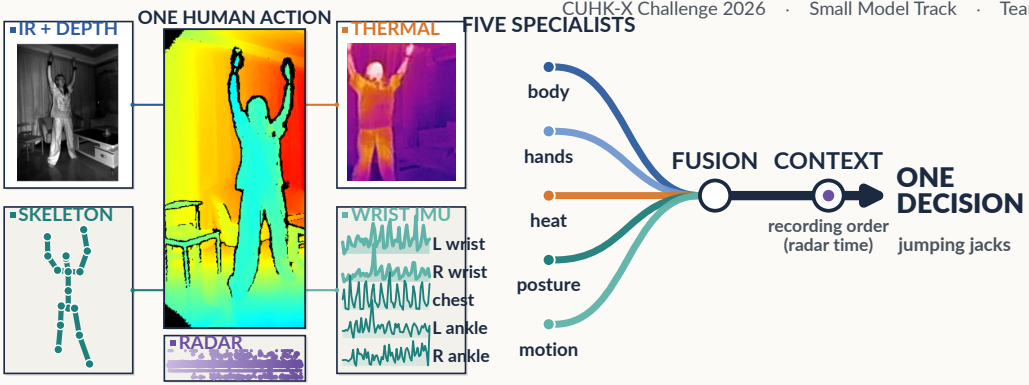
FIVE SENSORS.
ONE DECISION.

Can a machine recognise everyday activities
without an ordinary camera?

Multimodal activity recognition under a 100 MB constraint

CUHK-X Challenge 2026 · Small Model Track · Team AGX

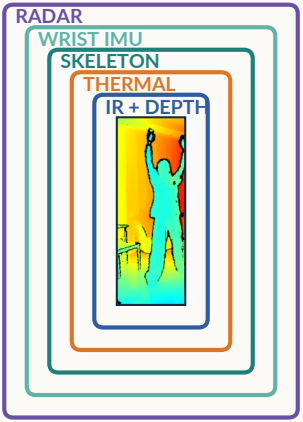


40	96.37 MB	0.877	UNSEEN
ACTIVITIES	ONE FILE	PRIVATE SCORE	PEOPLE AT TEST TIME

14th of 307 teams · 0 / 405 predictions changed across 5 tested GPU generations · no network at inference

WHAT IF THE CAMERA NEVER SAW YOUR FACE?

Recognise forty everyday activities from sensors that cannot identify anyone, for people the model has never seen, with the whole system in one file of at most 100 MB.





no ordinary camera

40 ACTIVITIES

18 TRAINING SUBJECTS

100 MB MAXIMUM, ONE FILE

Generalise to people the model has never seen.

2,933 training clips from 18 people; 405 test clips from 4 unseen people.

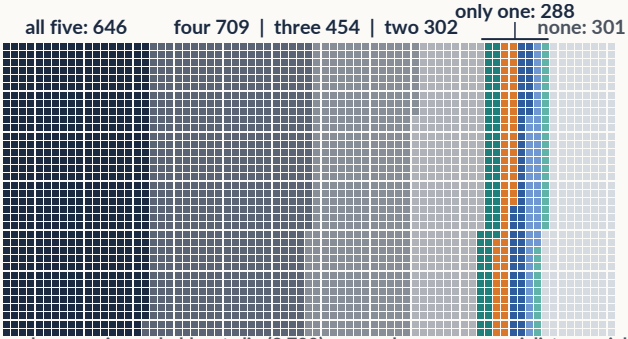
Infrared + depth, thermal, 17-joint skeleton, five inertial units, radar. No RGB camera anywhere in the data.

Scored on those new people, on reproducibility and on efficiency: cross-subject robustness is worth more than the leaderboard.

405 test clips · **unseen people** · **additional held-out subjects**

ONE SENSOR SEES ONLY PART OF THE STORY.

Every one of 2,700 held-out training clips, coloured by which specialists classify it correctly.



right for exactly one specialist

- **73** posture only
- **68** heat only
- **61** body only
- **52** hands only
- **34** motion only

nobody: 301 of 2,700

each square is one held-out clip (2,700): navy = how many specialists are right (five to two), a colour = exactly one, gray = none

2,399 / 2,700

correct for at least one specialist (0.889)

Different sensors make different mistakes.

REMOVE ANY ONE MEMBER

−11 TO −48 CLIPS

every specialist is load-bearing

FIVE SPECIALISTS. ONE DECISION.

Each specialist reads the stream it is built for; a prior-corrected log-linear sum fuses their votes; the recording's order adds context.



FIVE SPECIALISTS

body

hands

heat

posture

motion

0.781
fused, OOF

FUSION

$\sum_i w_i \log p_i$

0.830
decoded, OOF

CONTEXT

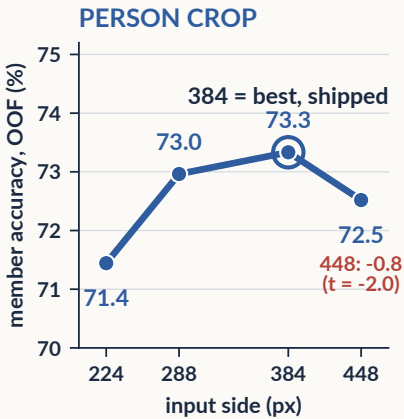
recording order
(radar time)

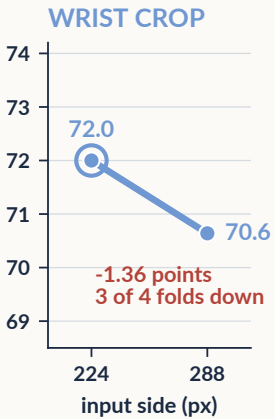
ONE DECISION

jumping jacks

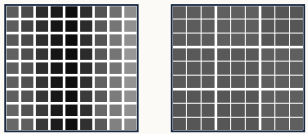
MORE PIXELS ARE NOT ALWAYS MORE INFORMATION.

Resolution helps the body crop up to 384 px and hurts the wrist crop already at 288 px; the difference is what one input pixel contains.





WHY: inside one input pixel



one source pixel = one input pixel

one source pixel = nine input pixels

person box 224 to 480 source px

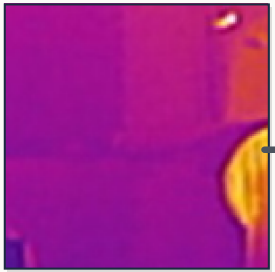
wrist box about 96 px

Resolution pays where the crop contains real detail.

THERMAL WAS USELESS. UNTIL WE READ IT CORRECTLY.

The same camera, three recipes: a 128 px person crop, the 128 px full frame, the 224 px full frame through the video recipe.

128 px crop, 3-D CNN



0.544

member accuracy, OOF

fusion gain: 0.00 at every weight

+2.4

128 px full frame, 3-D CNN



0.568

member accuracy, OOF

fusion gain: +0.83, not adopted

+5.6

points

4 of 4 folds

224 px full frame, MViTv2-S



0.627

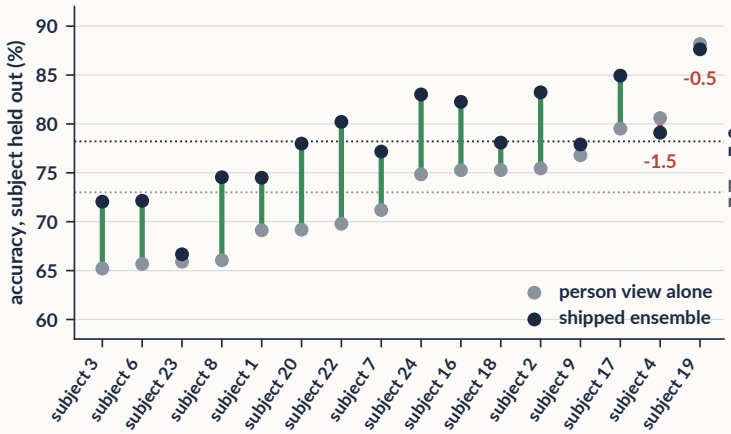
member accuracy, OOF

+46 decoded clips, +5 public clip

SAME SENSOR. DIFFERENT WAY OF READING IT.

THE REAL TEST IS THE PERSON YOU'VE NEVER SEEN.

Leave one training subject out, fit the rest, score the held-out person: the person view alone against the shipped ensemble, paired on 16 subjects.



73.0 → 78.2

mean accuracy over 16 held-out subjects

sd 6.4 → 5.4

the subjects become more alike

spread 23 → 21

points, hardest to easiest person

14 / 16 subjects improve

An 18-subject leave-one-subject-out run supports the direction; the organisers' additional held-out subjects remain the final external test.

WHAT HAPPENS WHEN A SENSOR DISAPPEARS?

The frozen fusion under two kinds of damage: a sensor absent, or one coefficient scaled by half to one and a half; every cell is the change in fused OOF clips.

one coefficient scaled, the others fixed

	ABSENT	x0.5	x0.75	x1	x1.25	x1.5
BODY	-48	-24	-7	0	+5	-7
HANDS	-25	-7	+2	0	0	-2
HEAT	-12	+4	+7	0	-2	-4
POSTURE	-23	-9	-5	0	-2	-9
MOTION	-11	-4	0	0	-2	-4

change in fused OOF clips (of 2,700) vs the shipped baseline 2,109
boxed column = the shipped setting

-48

CLIPS: worst single-sensor loss (-1.8 points, the body view)

-24 TO +7

all coefficient perturbations, ×0.5 to ×1.5

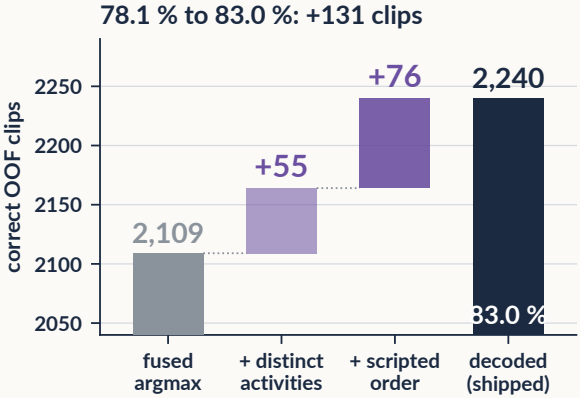
any re-weighting moves at most 8 of the

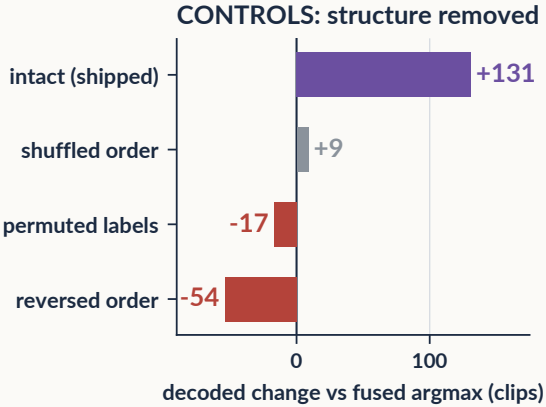
405 test rows

THE FUSION SITS ON A PLATEAU, NOT A PEAK.

THE RECORDING ITSELF KNOWS SOMETHING.

A recording session is a script of distinct activities in a fixed order; the decoder uses that order, learned from training metadata only, never from test labels.

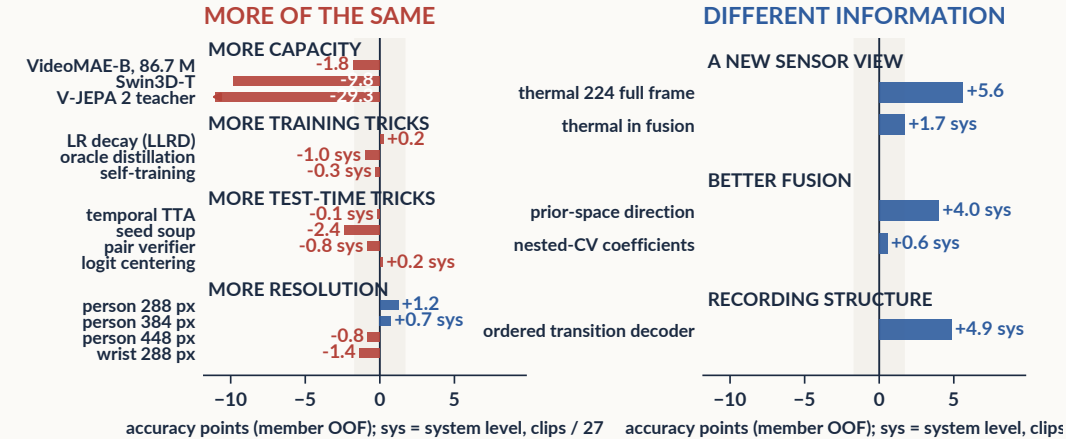




Label-free, and neutral without structure: shuffled +9, reversed -54, permuted labels -17.

SO WE TRIED THE OBVIOUS THINGS.

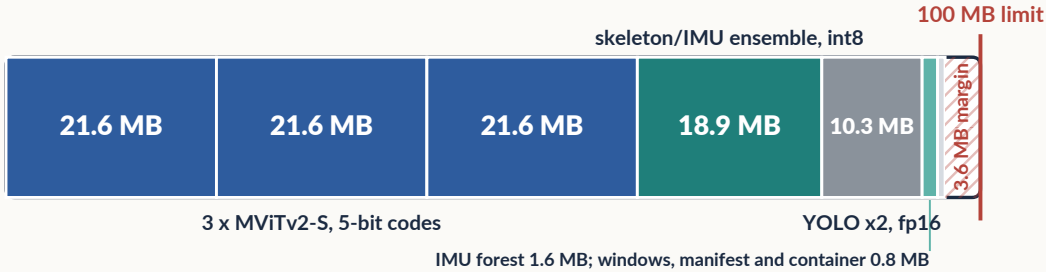
Every lever measured against the same bar: paired folds, two standard errors (seed sd 1.16 points). Red was rejected, blue adopted.



The bottleneck was not simply model capacity.

WE FIT AN ENTIRE SYSTEM INTO 100 MB.

The budget was an architectural constraint, not a compression exercise: which members to keep, at which precision, with both detectors inside.



64.8 + 18.9 + 10.3 + 1.6 + 0.8 = 96.37 MB: every inference-time weight, both detectors and the 405 crop windows, in one file

96.37 MB

every inference-time weight inside

Video 5-bit, ensemble int8, detectors fp16. **int8 detectors moved 150 of 405 crop windows**, so they stay fp16; int5 video cost one public clip against int6.

AND IT ACTUALLY RUNS.

One command, no network, data read-only, verified from the uploaded archives in a clean room, on 5 GPU generations.

ONE COMMAND

```
bash inference.sh
<data> <out.csv>
```

2 min 39 s

for all 405 clips on an A100; 4 min 53 s on a laptop RTX 4060; peak GPU memory 5.4 GB

GPU	posterior drift video / skeleton	predictions changed	CSV
RTX 4060	reference	0 / 405	IDENTICAL
A100	1.6e-3 / 4.4e-5	0 / 405	IDENTICAL
H100	2.0e-3 / 4.3e-5	0 / 405	IDENTICAL
H200	2.0e-3 / 4.3e-5	0 / 405	IDENTICAL
V100	1.7e-3 / 9.2e-5	0 / 405	IDENTICAL

NUMERICAL DRIFT EXISTS. THE DECISION DOES NOT MOVE.

0 / 405

predictions changed on 5 GPU generations; posteriors drift up to 2e-3

NETWORK OFF · READ-ONLY DATA · HOSTILE TESTS · CLEAN ROOM ×2

second selected submission byte-identical; primary reproduced 403/405 after the skeleton ensemble was pruned to fit 100 MB, the two decoder near-ties disclosed, not fitted (backup B8)

FIVE SENSORS.
ONE DECISION.

40	96.37 MB	0.877	UNSEEN
ACTIVITIES	ONE FILE	PRIVATE, 14th OF 307	PEOPLE

THE NEXT GAIN IS MORE SUBJECTS, NOT MORE MODEL.



Team AGX

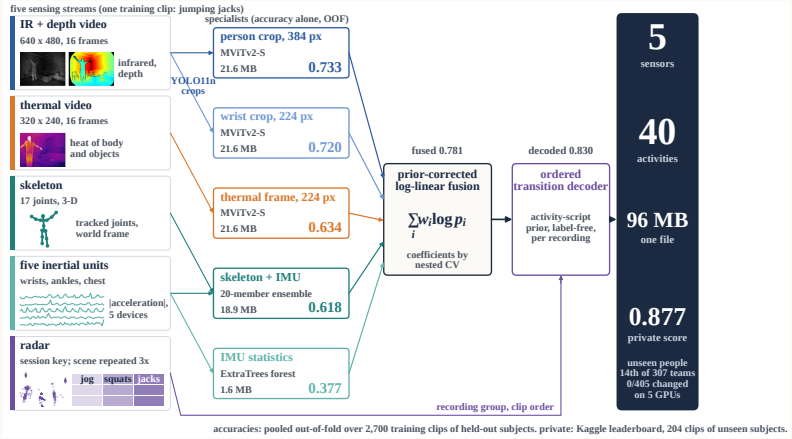
CUHK-X Challenge 2026 · Small Model Track · Atharv, BITS Pilani, K K Birla Goa Campus

BACKUP

architecture · fusion · members · the full landscape · subjects · decoder · package · limitations and disclosure

B1 THE SYSTEM, IN FULL

Five streams, five specialists, one file (report Fig. 1). Member accuracies are pooled OOF over 2,700 clips of held-out subjects; the card's 0.877 is the Kaggle private score.



B2 FUSION: THE RIGHT PROBABILITY SPACE

Members are combined as $\sum_i w_i \log p_i$. Video members are trained with a prior-corrected loss and emit **uniform-prior** posteriors; the skeleton ensemble is trained with plain cross-entropy and emits **train-prior** posteriors, so it is divided by the training prior before the sum. The direction is measured: the wrong way cost 4 points on an early submission pair (0.587 vs 0.627).

coefficient	shipped
person (body)	0.236
wrist (hands)	0.181
thermal (heat)	0.170
skeleton/IMU (posture)	0.332
IMU forest (motion)	0.297
class prior	0.25

Coefficients are the centroid of the top-25 settings of a nested cross-validation over a 768-point grid, which reproduces the nested-CV estimate without the selection optimism of the argmax. The stability atlas (slide 8, report Fig. 7) shows the plateau: every coefficient $\times 0.5$ to $\times 1.5$ stays within -24 to +7 clips.

Fused OOF change under perturbation (clips of 2,700)

	person	wrist	thermal	skeleton	IMU forest	class prior
absent, constant posterior	-48	-29	-11	-27	-13	n/a
absent, removed from fusion	-48	-25	-12	-23	-11	n/a
one coefficient scaled, the others fixed (shipped values boxed):						
coefficient x0.5	-24	-7	+4	-9	-4	0
coefficient x0.75	-7	+2	+7	-5	0	+1
coefficient x1	0	0	0	0	0	0
coefficient x1.25	+5	0	-2	-2	-2	-1
coefficient x1.5	-7	-2	-4	-9	-4	+2

worst case with a sensor absent: -48 clips = -1.8 points;
any coefficient from x0.5 to x1.5: within -24 to +7 clips

B3 WHY FIVE MEMBERS

Report Fig. 2: all 2,700 OOF clips grouped by which members classify them correctly; leave-one-member-out on the same posteriors.

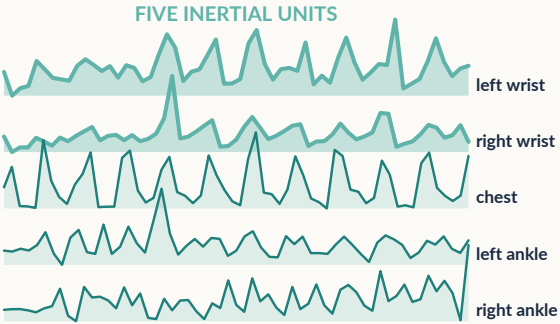


B3b THE SKELETON / IMU ENSEMBLE

What the posture specialist reads: the tracked skeleton and the five inertial units of the same clip.



17 joints in 3-D, frame 18 of the clip



|acceleration| over the six-second clip, one row per unit

20
MEMBERS

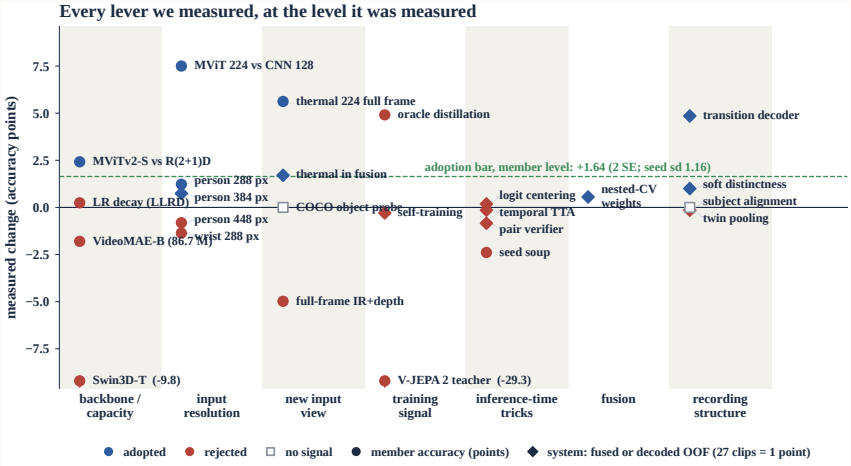
5
FAMILIES × 4 FOLDS

18.9 MB
INT8, ONE TENSOR SET

adaptive-graph ST-GCN and ST-GCN on joints; a temporal CNN on joint, velocity and bone features; two temporal CNNs on the inertial units (rotation-invariant features; gravity-aligned world frame, no absolute orientation)

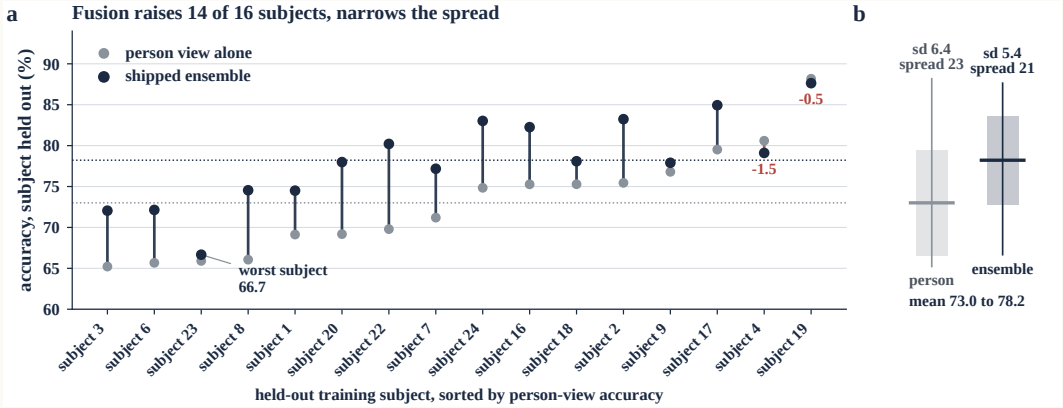
B4 THE FULL EXPERIMENT LANDSCAPE

Report Fig. 9: every lever on one axis at the level it was measured (circle = member points, diamond = system clips / 27), with the adoption band.



B5 SUBJECT-LEVEL DETAIL

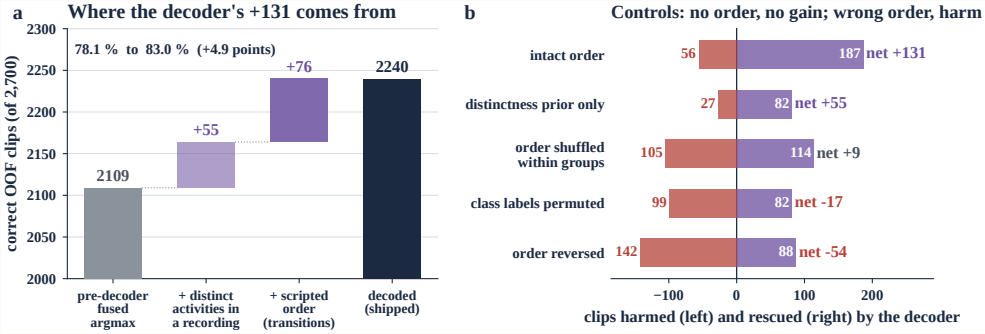
Report Fig. 6: the paired 16-subject comparison and the distributions (mean, ±1 sd, range).



Over all 18 subjects the person view has sd 6.9 and spread 28 (two subjects lack an ensemble value). Standard error of an 8-subject draw, which is what the private and held-out stages are: 2.4 → 1.9 points.

B6 DECODER CONTROLS

Report Fig. 8: where the decoder's +131 clips come from, and what happens when the structure is removed (rescued and harmed counted separately).



Settings: transition weight 0.5, conditional transition scores with unigram back-off, soft distinctness penalty 2.0, beam search per recording group; a clip without a radar key receives the fused argmax.

B7 THE PACKAGE

Drift exists; the predictions do not move

	max |dp| video	max |dp| skeleton	predictions changed	CSV bytes	network	run as
RTX 4060	ref.	ref.	**0 / 405**	identical	off	docker
A100	1.6e-03	4.4e-05	**0 / 405**	identical	off	native
H100	2.0e-03	4.3e-05	**0 / 405**	identical	off	apptainer
H200	2.0e-03	4.3e-05	**0 / 405**	identical	off	apptainer
V100	1.7e-03	9.2e-05	**0 / 405**	identical	off	apptainer

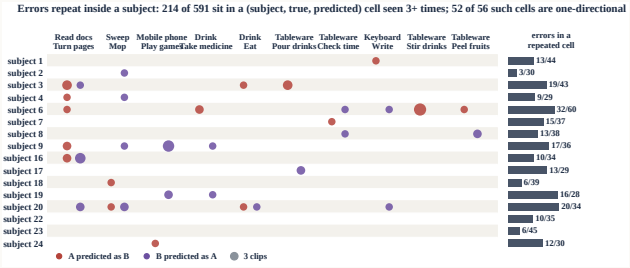
Archive `small_AGX_verification_v1.zip` (96.7 MB): `inference.sh`, `src/`, `artifacts/stage2_ship7.pth` (96.37 MB, sha256 recorded), `configs/` for both selected submissions, both CSVs byte-copied, README with the residual disclosure.

Image `ghcr.io/atharvgaur1845/cuhkx-small-agx:v1` (public) and `image.tar.gz` (3.59 GB), `pytorch/pytorch:2.7.1-cuda12.6`.

Gates G1 to G14: reproduction, offline, cache and id independence, synthetic hostile inputs (new ids, shuffled order, missing modalities, malformed files), docker with `--network none`, cross-GPU, content and secrets audit, clean-room from the uploaded archives with an independent comparer.

Timing 2 min 39 s (A100), 4 min 53 s (RTX 4060), peak GPU memory 5.4 GB.

B8 LIMITATIONS AND DISCLOSURE



Report Fig. 10: 214 of 591 fused errors sit in a (subject, true, predicted) cell seen three or more times; 52 of 56 such cells are one-directional.

The residual is subject-correlated. Twin recordings share the identical wrong label 68 % of the time; no lever that assumes independent evidence reaches it.

More model: measured, no gain. More subjects: the direction the evidence points to, not a quantified prediction (a frozen-feature probe saturates at six subjects).

Primary submission: 403/405. The skeleton ensemble was pruned 48 → 20 after that submission to meet 100 MB; two decoder near-ties flip (margins 0.012 and 0.006). About 900 coefficient settings tried, none closer. We ship the principled coefficients and disclose the rows; the second submission is byte-identical.

Decoder dependence: needs the recording key; neutral without it.